

Gene Set Enrichment Report

Generated by direct API calls to g:Profiler, Enrichr, and STRING. Species: *Mus musculus* (taxid 10090 / mmusculus). Raw tool output only – no interpretation applied.

Input gene list (n=25):
Slc30a9, Cd68, Rc3h1, Cnot9, Gsk3a, Gabarap, Rcan1, Itpr2, Jun, Lgals1, Lrrfip2, Gjb3, Atplb3, Mms19, Mical2, Zbtb34, Ddit3, Slc26a11, Hdac2, Mif, Gli3, Dst, Dtnbp1, Mafg, Tnfsf18

0. Simple summary (top 5 terms per source)

A flattened, easy-scan version of the detailed tables below. Same underlying numbers, fewer columns.

Table form

Source	Term	Adj. p-value	Genes
g:Profiler (GO:MF)	protein binding	9.80e-06	Slc30a9, Rc3h1, Cnot9, Gsk3a, Gabarap, Rcan1, Itpr2, Jun, Lgals1, Lrrfip2, Gjb3, Atplb3, Mms19, Mical2, Zbtb34, Hdac2, Mif, Gli3, Dst, Dtnbp1, Mafg, Tnfsf18
g:Profiler (GO:BP)	negative regulation of alpha-beta T cell differentiation	2.00e-05	Rc3h1, Lgals1, Gli3, Tnfsf18
g:Profiler (GO:BP)	negative regulation of alpha-beta T cell activation	1.18e-04	Rc3h1, Lgals1, Gli3, Tnfsf18
g:Profiler (GO:BP)	negative regulation of T cell differentiation	1.53e-04	Rc3h1, Lgals1, Gli3, Tnfsf18
g:Profiler (GO:BP)	positive regulation of macromolecule metabolic process	2.12e-04	Slc30a9, Rc3h1, Cnot9, Gsk3a, Gabarap, Jun, Mms19, Mical2, Hdac2, Mif, Gli3, Dtnbp1, Mafg
g:Profiler (GO:BP)	negative regulation of lymphocyte differentiation	2.46e-04	Rc3h1, Lgals1, Gli3, Tnfsf18
g:Profiler (GO:CC)	cytoplasm	1.75e-03	Slc30a9, Cd68, Rc3h1, Cnot9, Gsk3a, Gabarap, Rcan1, Itpr2, Jun, Lgals1, Gjb3, Atplb3, Mms19, Mical2, Slc26a11, Hdac2, Mif, Gli3, Dst, Dtnbp1
Enrichr – GO:BP	negative regulation of cellular macromolecule biosynthetic process (GO:2000113)	2.43e-02	JUN, GSK3A, HDAC2, DDIT3, CNOT9, GLI3
Enrichr – GO:BP	regulation of heart contraction (GO:0008016)	3.96e-02	GSK3A, ITPR2, ATP1B3
Enrichr – GO:BP	embryonic digit morphogenesis (GO:0042733)	3.96e-02	HDAC2, GLI3
Enrichr – GO:BP	positive regulation of nucleic acid-templated transcription (GO:1903508)	4.87e-02	JUN, HDAC2, DDIT3, MMS19, GLI3

Source	Term	Adj. p-value	Genes
Enrichr – GO:BP	positive regulation of transcription by RNA polymerase II (GO:0045944)	6.89e-02	SLC30A9, JUN, HDAC2, DDIT3, MICAL2, GLI3
Enrichr – GO:MF	LRR domain binding (GO:0030275)	1.18e-02	DDIT3, LRRFIP2
Enrichr – GO:MF	nuclear receptor coactivator activity (GO:0030374)	6.25e-02	SLC30A9, CNOT9
Enrichr – GO:MF	DNA-binding transcription factor binding (GO:0140297)	6.25e-02	JUN, HDAC2, DDIT3
Enrichr – GO:MF	histone deacetylase binding (GO:0042826)	7.42e-02	HDAC2, GLI3
Enrichr – GO:MF	cytokine receptor binding (GO:0005126)	7.42e-02	TNFSF18, MIF
Enrichr – GO:CC	intracellular membrane-bounded organelle (GO:0043231)	8.69e-02	SLC30A9, JUN, GSK3A, HDAC2, DST, DTNBP1, MICAL2, GLI3, RCAN1, GJB3, MAFG, DDIT3, MMS19, SLC26A11
Enrichr – GO:CC	nucleus (GO:0005634)	8.69e-02	RCAN1, SLC30A9, JUN, GSK3A, HDAC2, DST, MAFG, DDIT3, DTNBP1, MICAL2, MMS19, GLI3
Enrichr – GO:CC	P-body (GO:0000932)	8.69e-02	RC3H1, CNOT9
Enrichr – GO:CC	microtubule cytoskeleton (GO:0015630)	8.69e-02	DST, DTNBP1, MMS19
Enrichr – GO:CC	hemidesmosome (GO:0030056)	8.69e-02	DST
Enrichr – KEGG (mouse)	Kaposi sarcoma-associated herpesvirus infection	1.51e-02	RCAN1, JUN, ITPR2, GABARAP
Enrichr – KEGG (mouse)	Thyroid hormone signaling pathway	1.96e-02	RCAN1, HDAC2, ATP1B3
Enrichr – KEGG (mouse)	Apoptosis	1.96e-02	JUN, DDIT3, ITPR2
Enrichr – KEGG (mouse)	Non-alcoholic fatty liver disease (NAFLD)	1.96e-02	JUN, GSK3A, DDIT3
Enrichr – KEGG (mouse)	Oxytocin signaling pathway	1.96e-02	RCAN1, JUN, ITPR2
Enrichr – WikiPathways (mouse)	Calcium Regulation in the Cardiac Cell WP553	3.18e-02	GJB3, ITPR2, ATP1B3
Enrichr – WikiPathways (mouse)	IL-5 Signaling Pathway WP151	4.80e-02	JUN, GSK3A
Enrichr – WikiPathways (mouse)	Estrogen signaling WP1244	4.80e-02	JUN, HDAC2
Enrichr – WikiPathways (mouse)	Delta-Notch Signaling Pathway WP265	4.80e-02	JUN, HDAC2
Enrichr – WikiPathways (mouse)	Histone modifications WP300	4.99e-02	HDAC2
Enrichr – MGI Phenotype	abnormal olfactory nerve morphology MP:0005236	3.88e-02	LGALS1, GLI3
Enrichr – MGI Phenotype	embryonic lethality during organogenesis, incomplete penetrance MP:0011108	3.88e-02	JUN, HDAC2, GJB3, GLI3
Enrichr – MGI Phenotype	abnormal neural tube morphology MP:0002151	7.91e-02	JUN, ATP1B3, GLI3
Enrichr – MGI Phenotype	eyelids open at birth MP:0001302	7.91e-02	JUN, GLI3
Enrichr – MGI Phenotype	abnormal heart morphology MP:0000266	7.91e-02	RCAN1, JUN, HDAC2, GJB3

List form

g:Profiler – GO:MF

- **protein binding** — $p_{\text{adj}}=9.80\text{e-}06$ — genes: Slc30a9, Rc3h1, Cnot9, Gsk3a, Gabarap, Rcan1, Itpr2, Jun, Lgals1, Lrrfip2, Gjb3, Atp1b3, Mms19, Mical2, Zbtb34, Hdac2, Mif, Gli3, Dst, Dtnbp1, Mafg, Tnfsf18

g:Profiler — GO:BP

- **negative regulation of alpha-beta T cell differentiation** — $p_{\text{adj}}=2.00\text{e-}05$ — genes: Rc3h1, Lgals1, Gli3, Tnfsf18
- **negative regulation of alpha-beta T cell activation** — $p_{\text{adj}}=1.18\text{e-}04$ — genes: Rc3h1, Lgals1, Gli3, Tnfsf18
- **negative regulation of T cell differentiation** — $p_{\text{adj}}=1.53\text{e-}04$ — genes: Rc3h1, Lgals1, Gli3, Tnfsf18
- **positive regulation of macromolecule metabolic process** — $p_{\text{adj}}=2.12\text{e-}04$ — genes: Slc30a9, Rc3h1, Cnot9, Gsk3a, Gabarap, Jun, Mms19, Mical2, Hdac2, Mif, Gli3, Dtnbp1, Mafg
- **negative regulation of lymphocyte differentiation** — $p_{\text{adj}}=2.46\text{e-}04$ — genes: Rc3h1, Lgals1, Gli3, Tnfsf18

g:Profiler — GO:CC

- **cytoplasm** — $p_{\text{adj}}=1.75\text{e-}03$ — genes: Slc30a9, Cd68, Rc3h1, Cnot9, Gsk3a, Gabarap, Rcan1, Itpr2, Jun, Lgals1, Gjb3, Atp1b3, Mms19, Mical2, Slc26a11, Hdac2, Mif, Gli3, Dst, Dtnbp1

Enrichr — GO:BP

- **negative regulation of cellular macromolecule biosynthetic process (GO:2000113)** — $p_{\text{adj}}=2.43\text{e-}02$ — genes: JUN, GSK3A, HDAC2, DDIT3, CNOT9, GLI3
- **regulation of heart contraction (GO:0008016)** — $p_{\text{adj}}=3.96\text{e-}02$ — genes: GSK3A, ITPR2, ATP1B3
- **embryonic digit morphogenesis (GO:0042733)** — $p_{\text{adj}}=3.96\text{e-}02$ — genes: HDAC2, GLI3
- **positive regulation of nucleic acid-templated transcription (GO:1903508)** — $p_{\text{adj}}=4.87\text{e-}02$ — genes: JUN, HDAC2, DDIT3, MMS19, GLI3
- **positive regulation of transcription by RNA polymerase II (GO:0045944)** — $p_{\text{adj}}=6.89\text{e-}02$ — genes: SLC30A9, JUN, HDAC2, DDIT3, MICAL2, GLI3

Enrichr — GO:MF

- **LRR domain binding (GO:0030275)** — $p_{\text{adj}}=1.18\text{e-}02$ — genes: DDIT3, LRRFIP2

- nuclear receptor coactivator activity (GO:0030374) — p_{adj}=6.25e-02 — genes: SLC30A9, CNOT9
- DNA-binding transcription factor binding (GO:0140297) — p_{adj}=6.25e-02 — genes: JUN, HDAC2, DDIT3
- histone deacetylase binding (GO:0042826) — p_{adj}=7.42e-02 — genes: HDAC2, GLI3
- cytokine receptor binding (GO:0005126) — p_{adj}=7.42e-02 — genes: TNFSF18, MIF

Enrichr — GO:CC

- intracellular membrane-bounded organelle (GO:0043231) — p_{adj}=8.69e-02 — genes: SLC30A9, JUN, GSK3A, HDAC2, DST, DTNBP1, MICAL2, GLI3, RCAN1, GJB3, MAFG, DDIT3, MMS19, SLC26A11
- nucleus (GO:0005634) — p_{adj}=8.69e-02 — genes: RCAN1, SLC30A9, JUN, GSK3A, HDAC2, DST, MAFG, DDIT3, DTNBP1, MICAL2, MMS19, GLI3
- P-body (GO:0000932) — p_{adj}=8.69e-02 — genes: RC3H1, CNOT9
- microtubule cytoskeleton (GO:0015630) — p_{adj}=8.69e-02 — genes: DST, DTNBP1, MMS19
- hemidesmosome (GO:0030056) — p_{adj}=8.69e-02 — genes: DST

Enrichr — KEGG (mouse)

- Kaposi sarcoma-associated herpesvirus infection — p_{adj}=1.51e-02 — genes: RCAN1, JUN, ITPR2, GABARAP
- Thyroid hormone signaling pathway — p_{adj}=1.96e-02 — genes: RCAN1, HDAC2, ATP1B3
- Apoptosis — p_{adj}=1.96e-02 — genes: JUN, DDIT3, ITPR2
- Non-alcoholic fatty liver disease (NAFLD) — p_{adj}=1.96e-02 — genes: JUN, GSK3A, DDIT3
- Oxytocin signaling pathway — p_{adj}=1.96e-02 — genes: RCAN1, JUN, ITPR2

Enrichr — WikiPathways (mouse)

- Calcium Regulation in the Cardiac Cell WP553 — p_{adj}=3.18e-02 — genes: GJB3, ITPR2, ATP1B3
- IL-5 Signaling Pathway WP151 — p_{adj}=4.80e-02 — genes: JUN, GSK3A
- Estrogen signaling WP1244 — p_{adj}=4.80e-02 — genes: JUN, HDAC2
- Delta-Notch Signaling Pathway WP265 — p_{adj}=4.80e-02 — genes: JUN, HDAC2
- Histone modifications WP300 — p_{adj}=4.99e-02 — genes: HDAC2

Enrichr — MGI Phenotype

- **abnormal olfactory nerve morphology MP:0005236** — $p_{\text{adj}}=3.88e-02$ — genes: LGALS1, GLI3
 - **embryonic lethality during organogenesis, incomplete penetrance MP:0011108** — $p_{\text{adj}}=3.88e-02$ — genes: JUN, HDAC2, GJB3, GLI3
 - **abnormal neural tube morphology MP:0002151** — $p_{\text{adj}}=7.91e-02$ — genes: JUN, ATP1B3, GLI3
 - **eyelids open at birth MP:0001302** — $p_{\text{adj}}=7.91e-02$ — genes: JUN, GLI3
 - **abnormal heart morphology MP:0000266** — $p_{\text{adj}}=7.91e-02$ — genes: RCAN1, JUN, HDAC2, GJB3
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1. g:Profiler (g:GOSt)

Sources queried: GO:BP, GO:MF, GO:CC, KEGG, Reactome, WikiPathways. Correction method: g:SCS (default). "Adjusted p-value" below is g:Profiler's native corrected p_value output. **Note:** `Ddit3` maps ambiguously to two Ensembl mouse gene IDs (ENSMUSG00000025408 and ENSMUSG00000116429) and was excluded from this run's statistics as a result – effective query size 23–24 depending on term. All other 24 genes mapped uniquely.

Source	Term ID	Term name	Adj. p-value	Term size	Genes in overlap	Overlapping genes
GO:MF	GO:0005515	protein binding	9.804e-06	9549	22	Slc30a9, Rc3h1, Cnot9, Gsk3a, Gabarap, Rcan1, Itpr2, Jun, Lgals1, Lrrfp2, Gjb3, Atp1b3, Mms19, Mical2, Zbtb34, Hdac2, Mif, Gli3, Dst, Dtnbp1, Mafg, Tnfsf18
GO:BP	GO:0046639	negative regulation of alpha-beta T cell differentiation	2.004e-05	30	4	Rc3h1, Lgals1, Gli3, Tnfsf18
GO:BP	GO:0046636	negative regulation of alpha-beta T cell activation	1.182e-04	46	4	Rc3h1, Lgals1, Gli3, Tnfsf18
GO:BP	GO:0045581	negative regulation of T cell differentiation	1.532e-04	49	4	Rc3h1, Lgals1, Gli3, Tnfsf18
GO:BP	GO:0010604	positive regulation of macromolecule metabolic process	2.121e-04	2783	13	Slc30a9, Rc3h1, Cnot9, Gsk3a, Gabarap, Jun, Mms19, Mical2, Hdac2, Mif, Gli3, Dtnbp1, Mafg
GO:BP	GO:0045620	negative regulation of lymphocyte differentiation	2.456e-04	55	4	Rc3h1, Lgals1, Gli3, Tnfsf18
GO:BP	GO:2000320	negative regulation of T-helper 17 cell differentiation	5.146e-04	16	3	Rc3h1, Lgals1, Tnfsf18
GO:BP	GO:0009893	positive regulation of metabolic process	5.403e-04	3012	13	Slc30a9, Rc3h1, Cnot9, Gsk3a, Gabarap, Jun, Mms19, Mical2, Hdac2, Mif, Gli3, Dtnbp1, Mafg
GO:BP	GO:2000317	negative regulation of T-helper 17 type immune response	6.245e-04	17	3	Rc3h1, Lgals1, Tnfsf18
GO:BP	GO:0046637	regulation of alpha-beta T cell differentiation	7.333e-04	72	4	Rc3h1, Lgals1, Gli3, Tnfsf18
GO:BP	GO:0051254	positive regulation of RNA metabolic process	1.060e-03	1645	10	Slc30a9, Rc3h1, Cnot9, Gsk3a, Jun, Mms19, Mical2, Hdac2, Gli3, Mafg

Source	Term ID	Term name	Adj. p-value	Term size	Genes in overlap	Overlapping genes
GO:BP	GO:0048522	positive regulation of cellular process	1.301e-03	5332	16	Slc30a9, Rc3h1, Cnot9, Gsk3a, Gabarap, Jun, Lgals1, Atp1b3, Mms19, Mical2, Hdac2, Mif, Gli3, Dtnbp1, Mafg, Tnfsf18
GO:CC	GO:0005737	cytoplasm	1.746e-03	10727	20	Slc30a9, Cd68, Rc3h1, Cnot9, Gsk3a, Gabarap, Rcan1, Itpr2, Jun, Lgals1, Gjb3, Atp1b3, Mms19, Mical2, Slc26a11, Hdac2, Mif, Gli3, Dst, Dtnbp1
GO:BP	GO:0045623	negative regulation of T-helper cell differentiation	1.851e-03	24	3	Rc3h1, Lgals1, Tnfsf18
GO:BP	GO:0043371	negative regulation of CD4-positive, alpha-beta T cell differentiation	2.375e-03	26	3	Rc3h1, Lgals1, Tnfsf18
GO:BP	GO:2000319	regulation of T-helper 17 cell differentiation	2.670e-03	27	3	Rc3h1, Lgals1, Tnfsf18
GO:BP	GO:0010557	positive regulation of macromolecule biosynthetic process	3.047e-03	2340	11	Slc30a9, Cnot9, Gsk3a, Jun, Mms19, Mical2, Hdac2, Mif, Gli3, Dtnbp1, Mafg
GO:BP	GO:1902106	negative regulation of leukocyte differentiation	3.214e-03	104	4	Rc3h1, Lgals1, Gli3, Tnfsf18
GO:BP	GO:0048518	positive regulation of biological process	3.456e-03	5712	16	Slc30a9, Rc3h1, Cnot9, Gsk3a, Gabarap, Jun, Lgals1, Atp1b3, Mms19, Mical2, Hdac2, Mif, Gli3, Dtnbp1, Mafg, Tnfsf18
GO:BP	GO:1903707	negative regulation of hemopoiesis	4.021e-03	110	4	Rc3h1, Lgals1, Gli3, Tnfsf18

2. Enrichr

Gene list submitted as-is (mouse gene symbols); Enrichr internally uppercases symbols for matching against its libraries. Top 8 terms per library shown, ranked by adjusted p-value (Benjamini-Hochberg).

GO Biological Process (2021)

Term	p-value	Adjusted p-value	Odds ratio	Combined score	Overlapping genes
negative regulation of cellular macromolecule biosynthetic process (GO:2000113)	4.629e-05	2.426e-02	11.34	113.22	JUN, GSK3A, HDAC2, DDIT3, CNOT9, GLI3
regulation of heart contraction (GO:0008016)	2.079e-04	3.960e-02	30.13	255.45	GSK3A, ITPR2, ATP1B3
embryonic digit morphogenesis (GO:0042733)	2.267e-04	3.960e-02	108.47	910.29	HDAC2, GLI3
positive regulation of nucleic acid-templated transcription (GO:1903508)	3.714e-04	4.866e-02	9.62	75.97	JUN, HDAC2, DDIT3, MMS19, GLI3
positive regulation of transcription by RNA polymerase II (GO:0045944)	7.257e-04	6.892e-02	6.68	48.27	SLC30A9, JUN, HDAC2, DDIT3, MICAL2, GLI3
axonal transport (GO:0098930)	9.207e-04	6.892e-02	51.00	356.51	DST, DTNBP1
embryonic limb morphogenesis (GO:0030326)	9.207e-04	6.892e-02	51.00	356.51	HDAC2, GLI3
negative regulation of canonical Wnt signaling pathway (GO:0090090)	1.110e-03	7.163e-02	16.68	113.47	GSK3A, DDIT3, GLI3

GO Molecular Function (2021)

Term	p-value	Adjusted p-value	Odds ratio	Combined score	Overlapping genes
LRR domain binding (GO:0030275)	1.353e-04	1.177e-02	144.66	1288.68	DDIT3, LRRFIP2
nuclear receptor coactivator activity (GO:0030374)	1.988e-03	6.246e-02	33.97	211.32	SLC30A9, CNOT9
DNA-binding transcription factor binding (GO:0140297)	2.154e-03	6.246e-02	13.15	80.75	JUN, HDAC2, DDIT3
histone deacetylase binding (GO:0042826)	5.986e-03	7.415e-02	19.00	97.25	HDAC2, GLI3
cytokine receptor binding (GO:0005126)	7.570e-03	7.415e-02	16.78	81.93	TNFSF18, MIF

Term	p-value	Adjusted p-value	Odds ratio	Combined score	Overlapping genes
sequence-specific DNA binding (GO:0043565)	1.084e-02	7.415e-02	5.22	23.62	JUN, HDAC2, MAFG, GLI3
sequence-specific double-stranded DNA binding (GO:1990837)	1.111e-02	7.415e-02	5.18	23.33	JUN, MAFG, DDIT3, GLI3
cAMP response element binding protein binding (GO:0008140)	1.120e-02	7.415e-02	103.99	467.16	DDIT3

GO Cellular Component (2021)

Term	p-value	Adjusted p-value	Odds ratio	Combined score	Overlapping genes
intracellular membrane-bounded organelle (GO:0043231)	1.358e-03	8.692e-02	3.64	24.01	SLC30A9, JUN, GSK3A, HDAC2, DST, DTNBP1, MICAL2, GLI3, RCAN1, GJB3, MAFG, DDIT3, MMS19, SLC26A11
nucleus (GO:0005634)	4.242e-03	8.692e-02	3.20	17.48	RCAN1, SLC30A9, JUN, GSK3A, HDAC2, DST, MAFG, DDIT3, DTNBP1, MICAL2, MMS19, GLI3
P-body (GO:0000932)	4.465e-03	8.692e-02	22.18	120.03	RC3H1, CNOT9
microtubule cytoskeleton (GO:0015630)	7.888e-03	8.692e-02	8.17	39.55	DST, DTNBP1, MMS19
hemidesmosome (GO:0030056)	8.718e-03	8.692e-02	138.67	657.63	DST
ESC/E(Z) complex (GO:0035098)	9.958e-03	8.692e-02	118.86	547.86	HDAC2
platelet dense tubular network membrane (GO:0031095)	1.120e-02	8.692e-02	103.99	467.16	ITPR2
apical dendrite (GO:0097440)	1.243e-02	8.692e-02	92.44	405.55	GSK3A

KEGG (2019, Mouse)

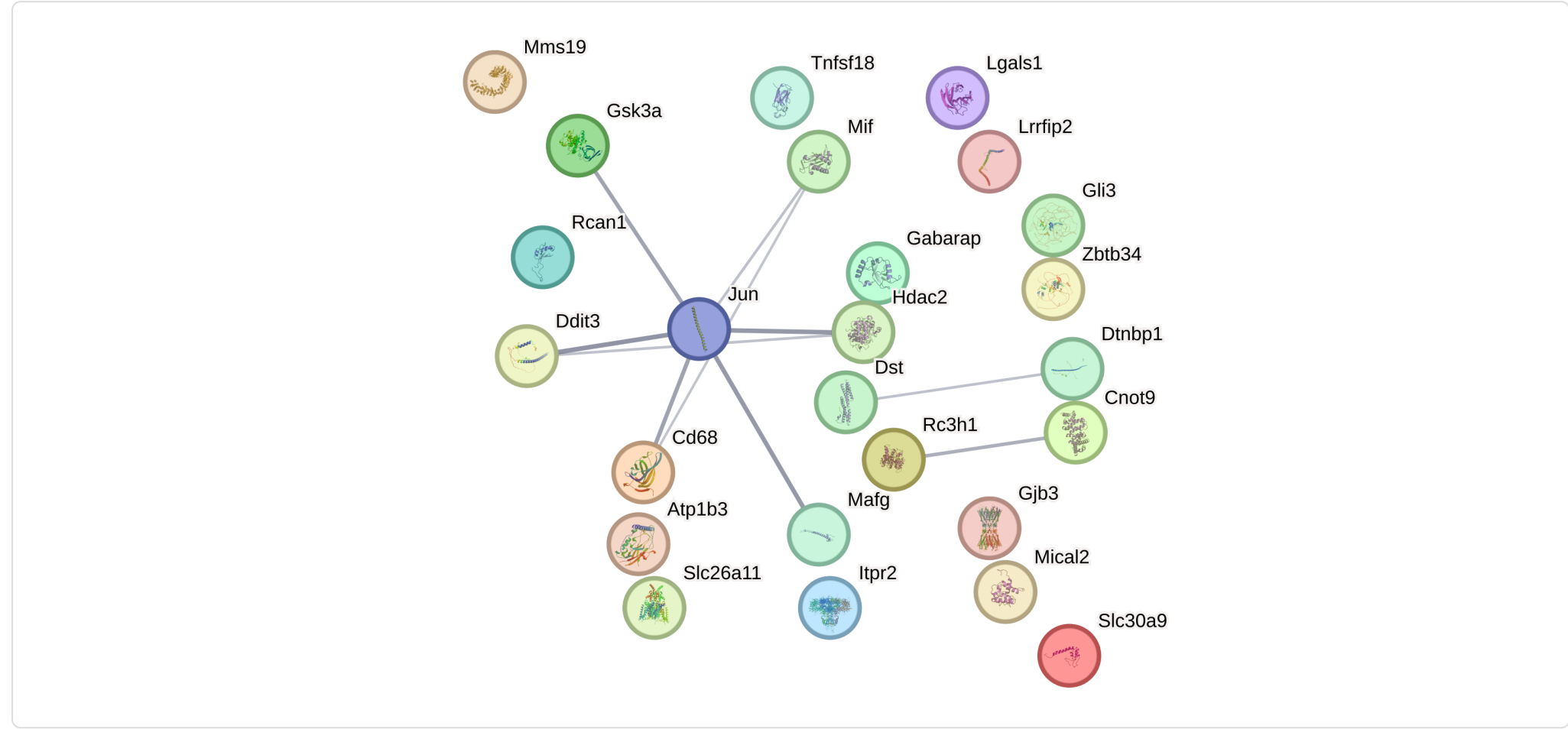
Term	p-value	Adjusted p-value	Odds ratio	Combined score	Overlapping genes
Kaposi sarcoma-associated herpesvirus infection	1.401e-04	1.513e-02	17.76	157.56	RCAN1, JUN, ITPR2, GABARAP
Thyroid hormone signaling pathway	3.884e-04	1.964e-02	24.18	189.93	RCAN1, HDAC2, ATP1B3
Apoptosis	7.041e-04	1.964e-02	19.60	142.28	JUN, DDIT3, ITPR2
Non-alcoholic fatty liver disease (NAFLD)	8.589e-04	1.964e-02	18.27	128.97	JUN, GSK3A, DDIT3
Oxytocin signaling pathway	9.092e-04	1.964e-02	17.90	125.37	RCAN1, JUN, ITPR2
NOD-like receptor signaling pathway	2.067e-03	3.461e-02	13.35	82.52	JUN, ITPR2, GABARAP
cAMP signaling pathway	2.243e-03	3.461e-02	12.96	79.05	JUN, ATP1B3, GLI3
Mitophagy	2.796e-03	3.774e-02	28.39	166.91	JUN, GABARAP

WikiPathways (2019, Mouse)

Term	p-value	Adjusted p-value	Odds ratio	Combined score	Overlapping genes
Calcium Regulation in the Cardiac Cell WP553	7.946e-04	3.179e-02	18.78	134.04	GJB3, ITPR2, ATP1B3
IL-5 Signaling Pathway WP151	3.343e-03	4.798e-02	25.84	147.30	JUN, GSK3A
Estrogen signaling WP1244	4.144e-03	4.798e-02	23.07	126.58	JUN, HDAC2
Delta-Notch Signaling Pathway WP265	4.798e-03	4.798e-02	21.36	114.04	JUN, HDAC2
Histone modifications WP300	6.235e-03	4.988e-02	208.03	1056.30	HDAC2
Adipogenesis genes WP447	1.208e-02	6.216e-02	13.07	57.72	DDIT3, MIF
Macrophage markers WP2271	1.243e-02	6.216e-02	92.44	405.55	CD68
Ethanol metabolism resulting in production of ROS by CYP2E1 WP4265	1.243e-02	6.216e-02	92.44	405.55	MAFG

MGI Mammalian Phenotype Level 4 (2021)

Term	p-value	Adjusted p-value	Odds ratio	Combined score	Overlapping genes
abnormal olfactory nerve morphology MP:0005236	1.781e-04	3.882e-02	123.98	1070.37	LGALS1, GLI3
embryonic lethality during organogenesis, incomplete penetrance MP:0011108	2.065e-04	3.882e-02	16.00	135.76	JUN, HDAC2, GJB3, GLI3
abnormal neural tube morphology MP:0002151	1.542e-03	7.907e-02	14.83	96.02	JUN, ATP1B3, GLI3
eyelids open at birth MP:0001302	1.914e-03	7.907e-02	34.65	216.87	JUN, GLI3
abnormal heart morphology MP:0000266	2.883e-03	7.907e-02	7.72	45.15	RCAN1, JUN, HDAC2, GJB3
abnormal myocardium layer morphology MP:0005329	4.465e-03	7.907e-02	22.18	120.03	RCAN1, HDAC2
decreased fibroblast proliferation MP:0011704	5.258e-03	7.907e-02	20.35	106.78	JUN, MIF
oxidative stress MP:0003674	6.625e-03	7.907e-02	18.01	90.34	RCAN1, JUN



3b. Interaction edges (returned by STRING)

Score columns are STRING's per-channel confidence scores (0–1): nscore=neighborhood, fscore=fusion, pscore=phylogenetic cooccurrence, ascore=coexpression, escore=experimental, dscore=database, tscore=textmining. "score" is the combined score.

Gene A	Gene B	Combined score	nscore	fscore	pscore	ascore	escore	dscore	tscore
Cd68	Mif	0.405	0	0	0	0.066	0	0	0.39
Cd68	Jun	0.642	0	0	0	0	0	0	0.642
Hdac2	Ddit3	0.424	0	0	0	0.047	0.132	0	0.36
Hdac2	Jun	0.707	0	0	0	0	0.096	0	0.69
Ddit3	Jun	0.722	0	0	0	0.077	0.123	0	0.685
Mif	Jun	0.43	0	0	0	0	0.065	0	0.415
Mafg	Jun	0.699	0	0	0	0.057	0	0	0.694
Gsk3a	Jun	0.627	0	0	0	0	0.247	0	0.525
Dtnbp1	Dst	0.424	0	0	0	0	0	0	0.425
Cnot9	Rc3h1	0.553	0	0	0	0	0.423	0	0.258

3c. Per-gene connectivity (degree in returned network)

Gene	Degree (# edges)
Slc30a9	0
Cd68	2
Rc3h1	1
Cnot9	1
Gsk3a	1
Gabarap	0
Rcan1	0
Itpr2	0
Jun	6
Lgals1	0
Lrrfip2	0
Gjb3	0
Atp1b3	0
Mms19	0
Mical2	0

Gene	Degree (# edges)
Zbtb34	0
Ddit3	2
Slc26a11	0
Hdac2	2
Mif	2
Gli3	0
Dst	1
Dtnbp1	1
Mafg	1
Tnfsf18	0

3d. STRING functional enrichment (top 15 by FDR, across all categories returned)

Category	Term ID	Description	# genes	# genes in background	p-value	FDR	Genes
Process	GO:0010604	Positive regulation of macromolecule metabolic process	16	3742	2.380e-07	3.800e-03	Gabarap,Hdac2,Ddit3,Lrrfip2,Mif,Mical2,Mafg,Gsk3a,Dtnbp1,Tnfsf18,Cnot9,Jun,Gli3,Slc30a9,Rc3h1,Mms19
Process	GO:0031325	Positive regulation of cellular metabolic process	15	3304	3.630e-07	3.800e-03	Gabarap,Hdac2,Ddit3,Mif,Mical2,Mafg,Gsk3a,Dtnbp1,Tnfsf18,Cnot9,Jun,Gli3,Slc30a9,Rc3h1,Mms19
Process	GO:0051173	Positive regulation of nitrogen compound metabolic process	15	3325	3.950e-07	3.800e-03	Gabarap,Hdac2,Ddit3,Mif,Mical2,Mafg,Gsk3a,Dtnbp1,Tnfsf18,Cnot9,Jun,Gli3,Slc30a9,Rc3h1,Mms19
COMPARTMENTS	GOCC:0005737	Cytoplasm	20	7282	2.310e-06	5.100e-03	Gabarap,Cd68,Hdac2,Ddit3,Atp1b3,Mif,Gjb3,Mical2,Itp2,Slc26a11,Rcan1,Gsk3a,Dtnbp1,Cnot9,Lgals1,Jun,Gli3,Rc3h1,Mms19,Dst
Process	GO:0031323	Regulation of cellular metabolic process	18	5853	3.220e-06	1.020e-02	Gabarap,Hdac2,Ddit3,Lrrfip2,Mif,Mical2,Mafg,Rcan1,Gsk3a,Dtnbp1,Tnfsf18,Cnot9,Jun,Gli3,Zbtb34,Slc30a9,Rc3h1,Mms19
Process	GO:0046639	Negative regulation of alpha-beta T cell differentiation	3	28	5.810e-06	1.020e-02	Tnfsf18,Gli3,Rc3h1
Process	GO:0048583	Regulation of response to stimulus	15	4029	4.900e-06	1.020e-02	Gabarap,Hdac2,Ddit3,Lrrfip2,Mif,Rcan1,Gsk3a,Dtnbp1,Tnfsf18,Cnot9,Lgals1,Jun,Gli3,Rc3h1,Mms19

Category	Term ID	Description	# genes	# genes in background	p-value	FDR	Genes
Process	GO:0051171	Regulation of nitrogen compound metabolic process	18	5973	4.420e-06	1.020e-02	Gabarap,Hdac2,Ddit3,Lrrfip2,Mif,Mical2,Mafg,Rcan1,Gsk3a,Dtnbp1,Tnfsf18,Cnot9,Jun,Gli3,Zbtb34,Slc30a9,Rc3h1,Mms19
Process	GO:0051254	Positive regulation of RNA metabolic process	11	1937	3.740e-06	1.020e-02	Hdac2,Ddit3,Mical2,Mafg,Gsk3a,Cnot9,Jun,Gli3,Slc30a9,Rc3h1,Mms19
Process	GO:0080090	Regulation of primary metabolic process	18	6158	7.090e-06	1.120e-02	Gabarap,Hdac2,Ddit3,Lrrfip2,Mif,Mical2,Mafg,Rcan1,Gsk3a,Dtnbp1,Tnfsf18,Cnot9,Jun,Gli3,Zbtb34,Slc30a9,Rc3h1,Mms19
Keyword	KW-0678	Repressor	6	557	3.310e-05	1.790e-02	Hdac2,Ddit3,Mafg,Cnot9,Gli3,Rc3h1
Process	GO:0010468	Regulation of gene expression	16	5140	1.950e-05	1.910e-02	Hdac2,Ddit3,Lrrfip2,Mif,Mical2,Mafg,Gsk3a,Dtnbp1,Tnfsf18,Cnot9,Jun,Gli3,Zbtb34,Slc30a9,Rc3h1,Mms19
Process	GO:0045893	Positive regulation of transcription, DNA-templated	10	1791	1.450e-05	1.910e-02	Hdac2,Ddit3,Mical2,Mafg,Gsk3a,Cnot9,Jun,Gli3,Slc30a9,Mms19
Process	GO:0048518	Positive regulation of biological process	18	6612	2.110e-05	1.910e-02	Gabarap,Hdac2,Ddit3,Atp1b3,Lrrfip2,Mif,Mical2,Mafg,Gsk3a,Dtnbp1,Tnfsf18,Cnot9,Lgals1,Jun,Gli3,Slc30a9,Rc3h1,Mms19
Process	GO:0048522	Positive regulation of cellular process	17	5948	2.560e-05	1.910e-02	Gabarap,Hdac2,Ddit3,Atp1b3,Mif,Mical2,Mafg,Gsk3a,Dtnbp1,Tnfsf18,Cnot9,Lgals1,Jun,Gli3,Slc30a9,Rc3h1,Mms19

Data provenance

g:Profiler: POST <https://biit.cs.ut.ee/gprofiler/api/gost/profile/>

Enrichr: POST <https://maayanlab.cloud/Enrichr/addList> + GET [/Enrichr/enrich](#)

STRING: GET https://string-db.org/api/tsv/get_string_ids, [/network](#), [/enrichment](#), [/highres_image/network](#)

All calls made live against each service; no cached or precomputed data used. Report generated 2026-08-11 15:28.